

## DPP-4

- Q.1 Cosine of an angle between the vectors  $\vec{a} + \vec{b}$  and  $\vec{a} - \vec{b}$  if  $|\vec{a}| = 2$ ,  $|\vec{b}| = 1$  and  $\vec{a} \wedge \vec{b} = 60^\circ$  is  
(A)  $\sqrt{3/7}$  (B)  $9/\sqrt{21}$  (C)  $3/\sqrt{7}$  (D) none
- Q.2 An arc AC of a circle subtends a right angle at the centre O. The point B divides the arc in the ratio 1 : 2. If  $\vec{OA} = \vec{a}$  &  $\vec{OB} = \vec{b}$ , then the vector  $\vec{OC}$  in terms of  $\vec{a}$  &  $\vec{b}$ , is  
(A)  $\sqrt{3}\vec{a} - 2\vec{b}$  (B)  $-\sqrt{3}\vec{a} + 2\vec{b}$  (C)  $2\vec{a} - \sqrt{3}\vec{b}$  (D)  $-2\vec{a} + \sqrt{3}\vec{b}$
- Q.3 For two particular vectors  $\vec{A}$  and  $\vec{B}$  it is known that  $\vec{A} \times \vec{B} = \vec{B} \times \vec{A}$ . What must be true about the two vectors?  
(A) At least one of the two vectors must be the zero vector.  
(B)  $\vec{A} \times \vec{B} = \vec{B} \times \vec{A}$  is true for any two vectors.  
(C) One of the two vectors is a scalar multiple of the other vector.  
(D) The two vectors must be perpendicular to each other.
- Q.4 'P' is a point inside the triangle ABC, such that  $BC\left(\vec{PA}\right) + CA\left(\vec{PB}\right) + AB\left(\vec{PC}\right) = 0$ , then for the triangle ABC the point P is its :  
(A) incentre (B) circumcentre (C) centroid (D) orthocentre
- Q.5 The vector equations of two lines  $L_1$  and  $L_2$  are respectively  
 $\vec{r} = 17\hat{i} - 9\hat{j} + 9\hat{k} + \lambda(3\hat{i} + \hat{j} + 5\hat{k})$  and  $\vec{r} = 15\hat{i} - 8\hat{j} - \hat{k} + \mu(4\hat{i} + 3\hat{j})$   
I  $L_1$  and  $L_2$  are skew lines  
II  $(11, -11, -1)$  is the point of intersection of  $L_1$  and  $L_2$   
III  $(-11, 11, 1)$  is the point of intersection of  $L_1$  and  $L_2$   
IV  $\cos^{-1}(3/\sqrt{35})$  is the acute angle between  $L_1$  and  $L_2$   
then, which of the following is true?  
(A) II and IV (B) I and IV (C) IV only (D) III and IV
- Q.6 Given three vectors  $\vec{a}$ ,  $\vec{b}$  &  $\vec{c}$  each two of which are non collinear. Further if  $(\vec{a} + \vec{b})$  is collinear with  $\vec{c}$ ,  $(\vec{b} + \vec{c})$  is collinear with  $\vec{a}$  &  $|\vec{a}| = |\vec{b}| = |\vec{c}| = \sqrt{2}$ . Then the value of  $\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a}$  :  
(A) is 3 (B) is -3 (C) is 0 (D) cannot be evaluated
- Q.7 For some non zero vector  $\vec{V}$ , if the sum of  $\vec{V}$  and the vector obtained from  $\vec{V}$  by rotating it by an angle  $2\alpha$  equals to the vector obtained from  $\vec{V}$  by rotating it by  $\alpha$  then the value of  $\alpha$ , is  
(A)  $2n\pi \pm \frac{\pi}{3}$  (B)  $n\pi \pm \frac{\pi}{3}$  (C)  $2n\pi \pm \frac{2\pi}{3}$  (D)  $n\pi \pm \frac{2\pi}{3}$   
where n is an integer.

- Q.8 Let  $\vec{u}, \vec{v}, \vec{w}$  be such that  $|\vec{u}| = 1, |\vec{v}| = 2, |\vec{w}| = 3$ . If the projection of  $\vec{v}$  along  $\vec{u}$  is equal to that of  $\vec{w}$  along  $\vec{u}$  and vectors  $\vec{v}, \vec{w}$  are perpendicular to each other then  $|\vec{u} - \vec{v} + \vec{w}|$  equals  
 (A) 2 (B)  $\sqrt{7}$  (C)  $\sqrt{14}$  (D) 14
- Q.9 If  $\vec{a}$  and  $\vec{b}$  are non zero, non collinear, and the linear combination  $(2x - y)\vec{a} + 4\vec{b} = 5\vec{a} + (x - 2y)\vec{b}$  holds for real x and y then x + y has the value equal to  
 (A) -3 (B) 1 (C) 17 (D) 3
- Q.10 In the isosceles triangle ABC  $|\vec{AB}| = |\vec{BC}| = 8$ , a point E divides AB internally in the ratio 1 : 3, then the cosine of the angle between  $\vec{CE}$  &  $\vec{CA}$  is (where  $|\vec{CA}| = 12$ )  
 (A)  $-\frac{3\sqrt{7}}{8}$  (B)  $\frac{3\sqrt{8}}{17}$  (C)  $\frac{3\sqrt{7}}{8}$  (D)  $-\frac{3\sqrt{8}}{17}$
- Q.11 If  $\vec{p} = 3\vec{a} - 5\vec{b}$ ;  $\vec{q} = 2\vec{a} + \vec{b}$ ;  $\vec{r} = \vec{a} + 4\vec{b}$ ;  $\vec{s} = -\vec{a} + \vec{b}$  are four vectors such that  $\sin(\vec{p} \wedge \vec{q}) = 1$  and  $\sin(\vec{r} \wedge \vec{s}) = 1$  then  $\cos(\vec{a} \wedge \vec{b})$  is :  
 (A)  $-\frac{19}{5\sqrt{43}}$  (B) 0 (C) 1 (D)  $\frac{19}{5\sqrt{43}}$
- Q.12 Given an equilateral triangle ABC with side length equal to 'a'. Let M and N be two points respectively on the side AB and AC such that  $\vec{AN} = K\vec{AC}$  and  $\vec{AM} = \frac{\vec{AB}}{3}$ . If  $\vec{BN}$  and  $\vec{CM}$  are orthogonal then the value of K is equal to  
 (A)  $\frac{1}{5}$  (B)  $\frac{1}{4}$  (C)  $\frac{1}{3}$  (D)  $\frac{1}{2}$